$\boxed{\star}$ - points awarded

ÖZYEĞİN UNIVERSITY
Math 103/113
Spring 2021

Final Exam Solutions

Q1) (Curve Sketching) [$6+6+4=16$ points]

Consider the function $f(x) = \cos x + \sin x$ restricted to the interval $0 \le x \le 2\pi$.

(a) Find the intervals where $f(x)$ is increasing / decreasing. Identify the coordinates of any local extreme points.

(b) Find the intervals where $f(x)$ is concave up / concave down. Identify the coordinates of any inflection points.

(c) Sketch the graph of $y = f(x)$. Indicate each important point on the graph. $\boxed{2}$

(a) $f'(x) = \cos x - \sin x = 0 \Rightarrow \begin{cases} \tan x = 1 \\ 0 \le x \le 2\pi \end{cases} \Rightarrow x = \frac{\pi}{4},\ x = \frac{5\pi}{4}$

	0		$\pi/4$		$5\pi/4$		2π
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$		$\nearrow$		$\searrow$		$\nearrow$	

f is increasing on $(0, \frac{\pi}{4}) \cup (\frac{5\pi}{4}, 2\pi)$.

f is decreasing on $(\frac{\pi}{4}, \frac{5\pi}{4})$. $\boxed{2}$

local maximum at $(\frac{\pi}{4}, \sqrt{2})$.

local minimum at $(\frac{5\pi}{4}, -\sqrt{2})$. $\boxed{2}$

(b) $f''(x) = -\sin x - \cos x = 0 \Rightarrow \begin{cases} \tan x = -1 \\ 0 \le x \le 2\pi \end{cases} \Rightarrow x = \frac{3\pi}{4},\ x = \frac{7\pi}{4}$ $\boxed{2}$

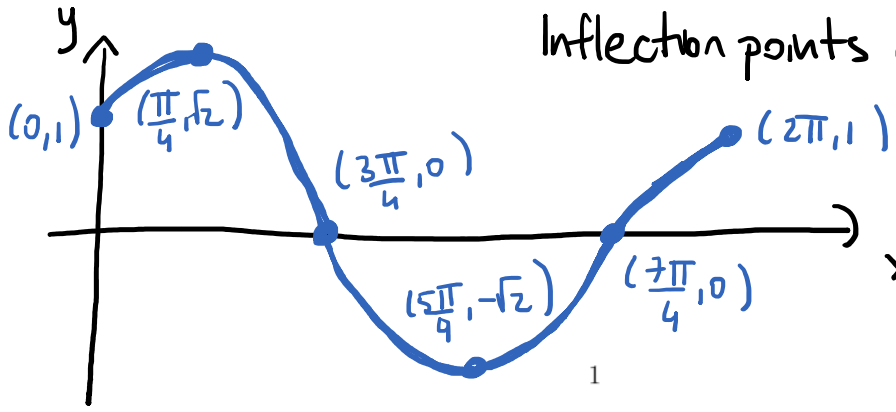
	0		$3\pi/4$		$7\pi/4$		2π
$f''(x)$		$-$	0	$+$	0	$-$	
$f(x)$		$\cap$		$\cup$		$\cap$	

f is concave up on $(\frac{3\pi}{4}, \frac{7\pi}{4})$ $\boxed{2}$

f is concave down on $(0, \frac{3\pi}{4}) \cup (\frac{7\pi}{4}, 2\pi)$.

Inflection points at $(\frac{3\pi}{4}, 0)$ and $(\frac{7\pi}{4}, 0)$ $\boxed{2}$

(c)



$\times$ $\boxed{4}$

Q2) (Indeterminate Forms and L'Hospital's Rule) [14 points]

Evaluate the limit $\lim_{x\to 0} \frac{\sin(x^2)}{x\sin x}$.

$$\lim_{x\to 0} \frac{\sin(x^2)}{x\sin x} = \lim_{x\to 0} \frac{2x\cos(x^2)}{\sin x + x\cos x}$$ L'Hosprtal $\boxed{6}$

$$= \lim_{x\to 0} \frac{2\cos(x^2) - 4x^2\sin(x^2)}{2\cos x - x\sin x}$$ L'Hosptal again $\boxed{6}$

$$= \frac{2\cos(0) - 4(0)\sin(0)}{2\cos(0) - (0)\sin(0)}$$

$$= \frac{2-0}{2-0}$$

$$= \boxed{1}$$

$\boxed{2}$

Q3) (The Fundamental Theorem of Calculus) [14 points]

Find $\frac{dy}{dx}$ for $y = x\int_2^{x^2} \sec t^3\, dt$.

$y = xF(x^2)$ where $F(x) = \int_2^x \sec t^3 dt$.

$\frac{dy}{dx} = \frac{d}{dx}\left[xF(x^2)\right]$

$= F(x^2) + x\frac{d}{dx}\left[F(x^2)\right]$ product rule $\boxed{4}$

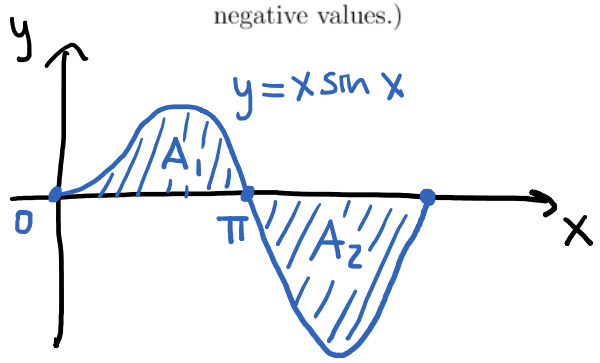
$= F(x^2) + 2x^2F'(x^2)$ chain rule $\boxed{4}$

$= \int_2^{x^2} \sec t^3 dt + 2x^2(\sec x^6)$ Fundamental Thm. part I

$\boxed{6}$

Q4) (Area Between Curves) [14 points]

Find the area of the region enclosed by the curve $y = x\sin x$ and the x-axis over the interval $0 \le x \le 2\pi$. (*Hint:* Be careful about where the function takes positive or negative values.)



$$A = A_1 + A_2$$

$$= \int_0^{\pi} x\sin x\,dx + \left|\int_{\pi}^{2\pi} x\sin x\,dx\right|$$

$\boxed{2}$

$$\int_0^{\pi} x\sin x\,dx = -x\cos x\Big|_0^{\pi} + \int_0^{\pi}\cos x\,dx$$

$\boxed{5}$

$$\begin{bmatrix} u = x, & dv = \sin x\,dx \\ du = dx, & v = -\cos x \end{bmatrix}$$

$$= -x\cos x\Big|_0^{\pi} + \sin x\Big|_0^{\pi}$$

$$= -\pi\cos\pi + 0\cos 0 + \sin\pi - \sin 0$$

$$= \pi$$

$$\int_{\pi}^{2\pi} x\sin x\,dx = -x\cos x\Big|_{\pi}^{2\pi} + \int_{\pi}^{2\pi}\cos x\,dx$$

$\boxed{5}$

$$= -x\cos x\Big|_{\pi}^{2\pi} + \sin x\Big|_{\pi}^{2\pi}$$

$$= -2\pi\cos 2\pi + \pi\cos\pi + \sin 2\pi - \sin\pi$$

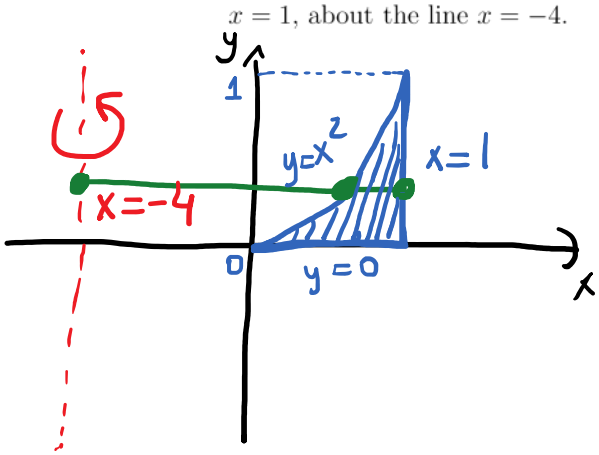
$$= -3\pi$$

$$A = \pi + |-3\pi| = \boxed{4\pi}$$

$\boxed{2}$

Q5) (Volumes Using Cross-Sections) [14 points]

Find the volume of the solid generated by revolving the region in the first quadrant bounded above by the curve $y = x^2$, below by the x-axis, and on the right by the line $x = 1$, about the line $x = -4$.



$$V = \int_0^1 \left(\pi [R(y)]^2 - \pi [r(y)]^2\right) dy$$

$$R(y) = 5, \quad r(y) = 4 + \sqrt{y}$$

$$V = \pi \int_0^1 \left([5]^2 - [4+\sqrt{y}]^2\right) dy \qquad \boxed{8}$$

$$= \pi \int_0^1 \left(9 - 8\sqrt{y} - y\right) dy$$

$$= \pi \left[9y - \frac{16}{3} y^{3/2} - \frac{1}{2} y^2\right]_0^1 \qquad \boxed{6}$$

$$= \pi \left[\left(9 - \frac{16}{3} - \frac{1}{2}\right) - 0\right]$$

$$= \boxed{\frac{19}{6}\pi}$$

Q6) (Techniques of Integration) [14 points]

Evaluate the integral $\int \frac{2x^4+6x^2-1}{x^3+3x}\,dx$.

$$\frac{2x^4+6x^2-1}{x^3+3x} = 2x - \frac{1}{x(x^2+3)}$$ $\boxed{2}$

$$\frac{1}{x(x^2+3)} = \frac{A}{x} + \frac{Bx+C}{x^2+3}$$

$$1 = A(x^2+3) + (Bx+C)x = (A+B)x^2 + Cx + 3A$$

$$A+B=0,\ C=0,\ 3A=1 \Rightarrow A=\frac{1}{3},\ B=\frac{-1}{3},\ C=0$$ $\boxed{6}$

$$\frac{2x^4+6x^2-1}{x^3+x} = 2x - \frac{1}{3x} + \frac{x}{3(x^2+3)}$$

$$\int \frac{2x^4+6x^2-1}{x^3+x}\,dx = \int 2x\,dx - \frac{1}{3}\int \frac{1}{x}\,dx + \frac{1}{3}\int \frac{x}{x^2+3}\,dx$$

$$= \boxed{x^2 - \frac{1}{3}\ln|x| + \frac{1}{6}\ln(x^2+3) + C}$$

$\boxed{6}$

Q7) (Improper Integrals) [14 points]

Evaluate the improper integral $\int_{-\infty}^{\infty} \frac{4\,dx}{x^2+16}$.

$$\int_{-\infty}^{\infty} \frac{4\,dx}{x^2+16} = \int_{-\infty}^{0} \frac{4\,dx}{x^2+16} + \int_{0}^{\infty} \frac{4\,dx}{x^2+16} \qquad \boxed{2}$$

$$\int_{0}^{\infty} \frac{4\,dx}{x^2+16} = \lim_{b\to\infty} \int_{0}^{b} \frac{4\,dx}{x^2+16}$$

$$= \lim_{b\to\infty} \left(\arctan\left(\frac{x}{4}\right) \Big|_{0}^{b} \right)$$

$$= \lim_{b\to\infty} \left(\arctan\left(\frac{b}{4}\right) - \arctan 0 \right)$$

$$= \frac{\pi}{2} - 0 = \frac{\pi}{2} \qquad \boxed{5}$$

$$\int_{-\infty}^{0} \frac{4\,dx}{x^2+16} = \lim_{a\to-\infty} \int_{a}^{0} \frac{4\,dx}{x^2+16}$$

$$= \lim_{a\to-\infty} \left(\arctan\left(\frac{x}{4}\right) \Big|_{a}^{0} \right)$$

$$= \lim_{a\to-\infty} \left(\arctan 0 - \arctan\left(\frac{a}{4}\right) \right)$$

$$= 0 - \left(-\frac{\pi}{2}\right) = \frac{\pi}{2} \qquad \boxed{5}$$

$$\int_{-\infty}^{\infty} \frac{4\,dx}{x^2+16} = \frac{\pi}{2} + \frac{\pi}{2} = \boxed{\pi} \qquad \boxed{2}$$